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GEOPHYSICS

A Newly Determined Paleomagnetic Pole for the Late Ordovician in Siberia and Kinematics of the Siberian Platform on the Time Interval of ~505–455 Ma Ago

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Abstract—A paleomagnetic study of the Chertovskii Horizon of the Moyero River valley was carried out. Based on published and newly obtained data, the sequence of seven reliable determinations of the paleomagnetic pole position is obtained for Siberia within the time interval of ~505–450 Ma. This sequence makes it possible to determine with high precision the paleogeographic position of the Siberian platform over a period of ~50 Ma from the Middle Cambrian until the middle of the Upper Ordovician.

Keywords: Siberian Platform, Ordovician, Paleomagnetism, paleoreconstruction

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INTRODUCTION

Due to their lithological characteristics, relatively great thickness, excellent exposure, and good biostratigraphic coverage, the Cambrian and Ordovician sedimentary sections of the Siberian Platform represent a unique object for obtaining a detailed paleomagnetic record, characterizing, in particular, the paleogeographic position and horizontal movements of the Siberian Craton during the Early Paleozoic. These data are necessary for paleotectonic and paleogeographic reconstructions and for exploring the geodynamic evolution of the Siberian platform as well as of the entire crust of the Earth.

Long-term studies of the paleomagnetism of the Cambrian and Ordovician have already made it possible to identify reliably a number of paleomagnetic pole positions with good statistical substantiation, which are being used widely to solve these problems. Meanwhile, the paleomagnetic data available for the Lower Paleozoic are incomplete. Until now, there has been uncertainty about the position of the paleomagnetic pole of Siberia for the Lower Cambrian [1]. The paleomagnetic data for the Upper Ordovician is utterly insufficient: there are only two Siberian pole determinations, performed for the Dolborskii horizon of the Moyero River section [2] and for the Chertovskii-Baksan interval of the Ordovician section of the middle reaches of the Angara River [3].

Field studies of the section of the Moyero River (north of the Siberian Platform), carried out in the summer of 2020 included, among other things, sampling of the deposits of the Chertovskii horizon, corresponding in its stratigraphic position to the lower half of the Caradoc of the British Stratigraphic Scale and to the Lower Sandbian of the International Stratigraphic Scale. This contribution presents the results of paleomagnetic laboratory studies of a sample collection from the Chertovskii horizon, which made it possible to obtain a new paleomagnetic pole position for the Upper Ordovician of the Siberian Platform and propose a sequence of paleomagnetic poles describing the paleogeographic position and horizontal displacements of the Siberian Platform in the time interval of ~505–455 Ma.

BRIEF DESCRIPTION OF THE SECTION AND SAMPLING AND RESEARCH PROCEDURES

The study area is located on the southwestern slope of the Anabar Uplift, its southern and southeastern margins partly cover the northern wing of the Tunguska Syncline. The Paleozoic strata occur here almost horizontally and have only a slight dip towards the west-southwest, rarely exceeding 2°–3°. In some

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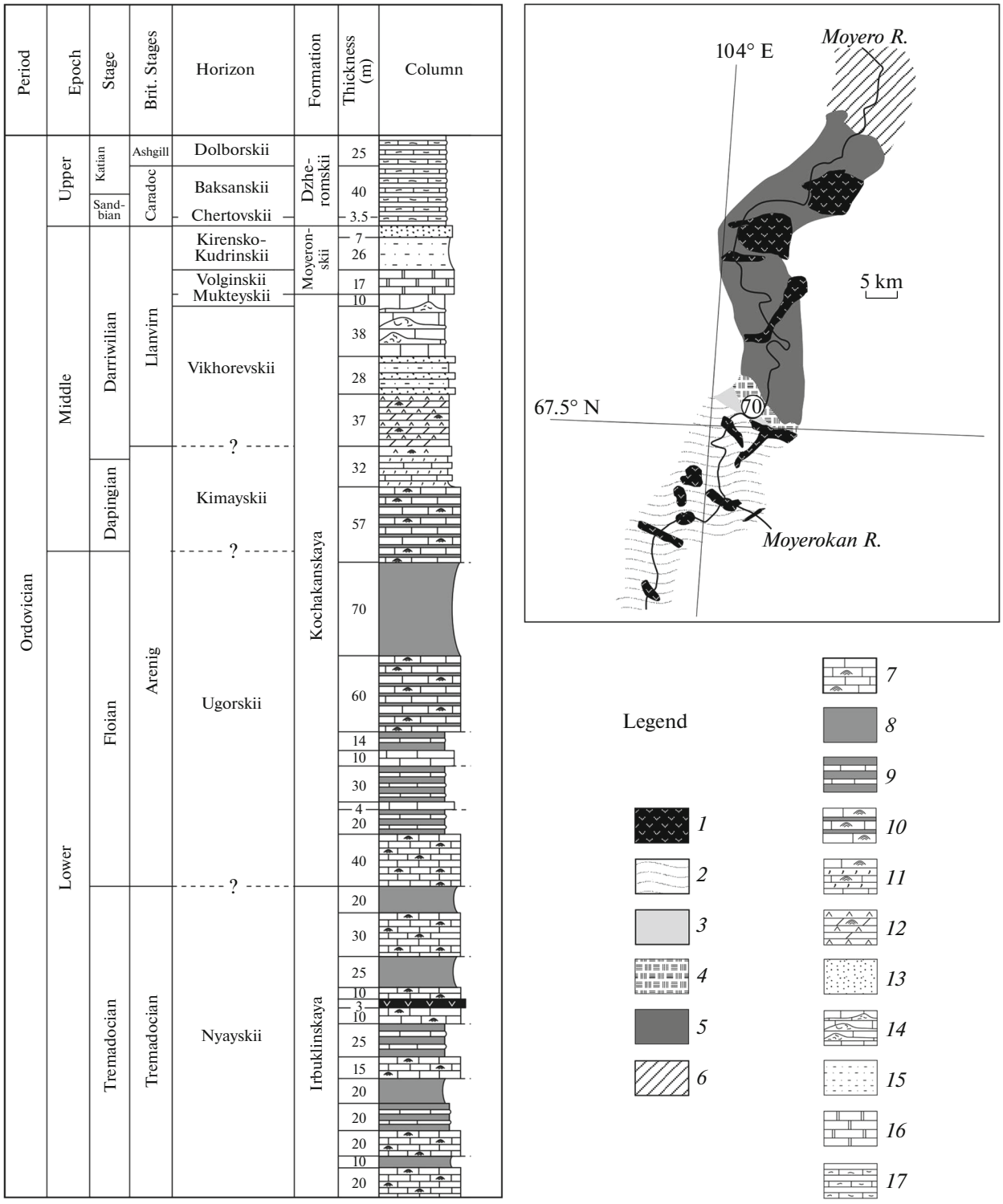


Fig. 1. Geographic and stratigraphic position of the studied section. (*1*) traps; (*2*) Silurian; (*3*) Upper Ordovician; (*4*) Middle Ordovician; (*5*) Lower Ordovician; (*6*) Cambrian; (*7*) predominantly gray-colored strata of warm-water oolitic limestones with stromatolites; (*8*) predominantly red-colored siltstones with crystal molds over halite and, sometimes, interlayers of gypsum; (*9*) alternation of red-colored siltstones and gray-colored, mainly oolitic limestones; (*10*) the same with stromatolite interbeds in the carbonate layers; (*11*) alternation of gray oolitic limestones with stromatolites and greenish gray siltstones with glauconite; (*12*) alternation beds of gypsum and dolomite with stromatolite; (*13*) quartz sandstones; (*14*) limestones and clayey limestones with *Angarella* bioherms; (*15*) violet and burgundy–red siltstones and fine-grained sandstones; (*16*) gray bioclastic limestones and limestone conglomerates; (*17*) cold-water carbonates, alternation of greenish gray siltstones and bioclastic limestones. A circle with the number 70 indicates the location of the outcrop studied.

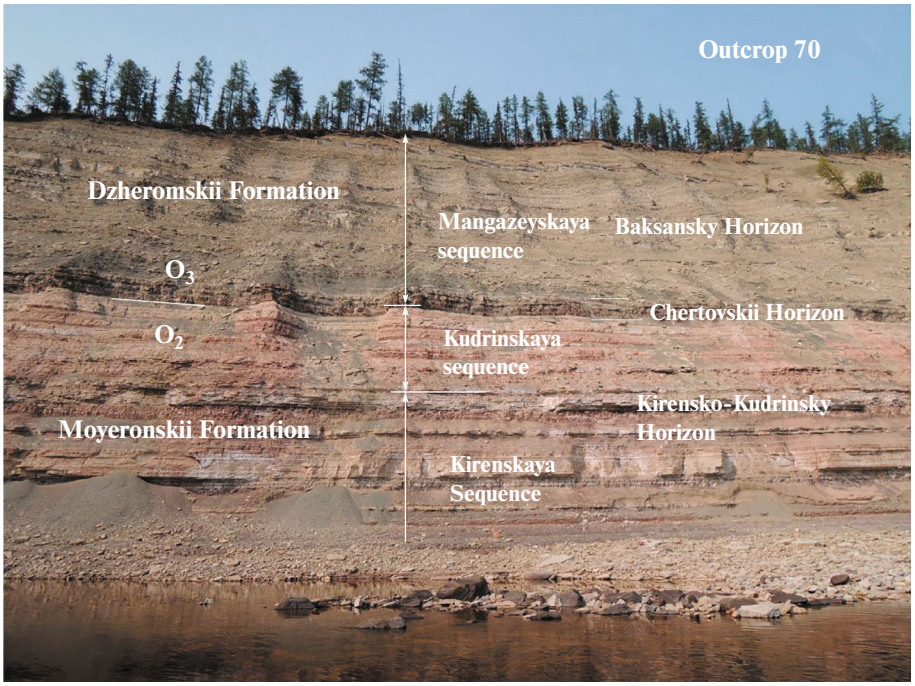


Fig. 2. Photo of outcrop 70 with stratigraphic boundaries marked.

places the sediments studied contain small local dislocations, usually associated with the intrusion of Permian–Triassic trap sills and dikes of various thicknesses. Biostratigraphic data based on the study of conodonts and macrofossils (brachiopods, cephalopods, etc.) indicate that the Ordovician section of the Moyero contains ten horizons out of 12 identified on the Siberian platform (Fig. 1). The latter suggests that the Moyero section is one of the most complete stratigraphic sections of the Ordovician of Northern Eurasia, and perhaps the entire planet.

The initial detailed description of the Ordovician reference section along the Moyero River was given in the monograph by Myagkova et al. [4]. Modern data on the stratigraphy, lithology, and formation conditions of the Ordovician deposits of the Siberian platform in general and the section along the Moyero River in particular were reported in [5–7]. The correlation of regional units (horizons) of the Ordovician of the Siberian Platform with the International Stratigraphic Scale was given in [8].

Here we give only a brief description of outcrop 70, where the rocks of the Chertovskii Horizon are exposed. The outcrop is located on the left bank of the Moyero River, 3.5 km above the mouth of the Bugarikty River, near the mouth of a small stream. The deposits of the Moyeronskii (Middle Ordovician) and Dzheromskii (Upper Ordovician) formations are exposed in the outcrop on a steep slope up to 80 m high (Figs. 1, 2).

The Moyeronskii Formation in outcrop 70 is represented by its upper subformation. Its lower part is composed of alternating beds of violet–red marls and beds of gray and yellowish gray bioclastic limestones with an admixture of quartz sand and/or quartz sandstones, and the upper part is represented by a layer of quartz sandstones with a thin slant and horizontal layering. The subformation is correlated with the Kirensko-Kudrinskii Horizon; it is about 30 m thick.

The Dzheromskii Formation comprises the Chertovskii, Baksanskii, and Dolborski horizons and corresponds to the Sandbian–Lower Katian stage of the Upper Ordovician [6]. In outcrop 70, the Dzheromskii Formation is composed mainly of typical cold-water carbonates.

Table 1. Direction of the characteristic magnetization in rocks of the Chertovskii Horizon of the Moyero River valley. *D*, declination; *I*, inclination; *K*, accuracy; α95, radius of the confidence circle (at the 95% level); *N*, the number of points in the Zijderveld diagrams, used for the calculation. Plat is the latitude of the pole, Plong is the longitude of the pole, and A95 is the radius of the confidence circle

	N	*D*, deg	*I*, deg	*K*	α95, deg
Forward polarity	15	141.4	10.5	31.3	7.0
Reverse polarity	8	311.7	13.0	23.7	11.6
Average direction	23	137.3	1.5	17.1	7.4
Average direction with accounting for remagnetizing circles	28	138.5	1.9	138.5	6.8

Paleomagnetic pole calculated (only by endpoints) for the characteristic component: Plat = –15.7°; Plong = 148.6°; A95 = 5.7°.

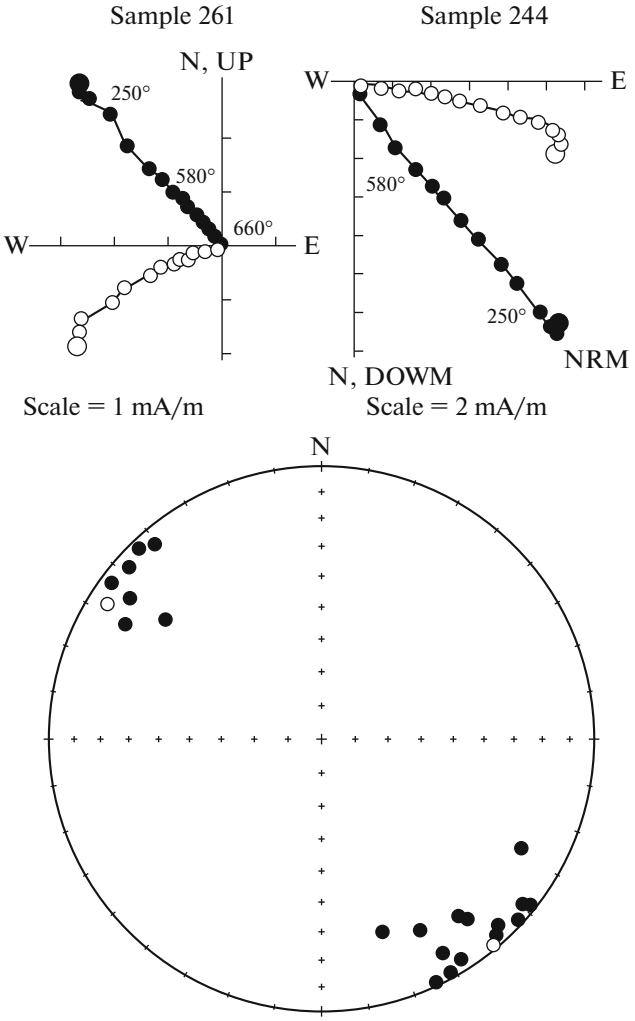


Fig. 3. Paleomagnetic record in samples of the Ordovician rocks from the Chertovskii Horizon, outcrop 70, Moiero River. Filled (not filled) circles on the plots denote the projections of the vectors of the characteristic component of magnetization onto the horizontal (vertical) plane. Filled (not filled) circles on the stereogram denote the projection of vectors onto the lower (upper) hemisphere. Plots and a stereogram are shown in ancient coordinate systems.

The lower boundary of the formation, which is also the lower boundary of the Chertovskii Horizon, is unconformable, lithologically expressed as an abrupt change of light gray fine-grained quartz sandstones crowning the Moyeronskii Formation to cherry–brown siltstones and greenish gray bioclastic limestones. This boundary is interpreted as the flood surface and the lower boundary of the Mangazeiskaya sedimentary sequence [6]. The Chertovskii Horizon is distinguished by a darker, reddish color and forms a brow in the relief (see Fig. 2). Its lower part is composed by a thin alternation of cherry–brown siltstones and greenish gray bioclastic limestones. The upper part of the Chertovskii Horizon is formed by greenish gray siltstones with a few interlayers and nodules of pinkish gray packstones. Deposits of the Chertovskii Horizon (2.5 m thick) represent the deepest part of the Mangazeiskaya sedimentary sequence and are interpreted as a transgressive system tract [6].

The middle and upper parts of the Dzheromskii Formation (Baksanskii and Dollyuorskii horizons) are composed of alternating gray and greenish gray organogenic–detrital, less often clayey, limestones and mudstones.

Forty samples were collected from the section of the Chertovskii Horizon in outcrop 70 with a sampling step of 5–7 cm for paleomagnetic analysis. The samples underwent laboratory processing in accordance with the standard procedure [9] in the laboratory of the Main Magnetic Field and Petromagnetism of the Institute of Physics of the Earth, Russian Academy of Sciences, using modern laboratory equipment (http://paleomag-ifz.ru/ru/equipment). Magnetic cleaning was carried out using a TD80 thermodemagnetizer (manufactured by Magnetic Measuremenets Ltd., Great Britain) in a special room shielded from an external magnetic field; the remanent magnetization was measured in the same room using a cryogenic magnetometer manufactured by 2GEnterprise (United States). In total, depending on the nature of the paleomagnetic recording, from 12 to 18 cleaning steps were performed up to temperatures corresponding to the hematite Curie point (~670–680°C).

LABORATORY PALEOMAGNETIC STUDY

The values of the magnetic susceptibility and natural remanent magnetization (NRM) of the rocks studied varied from 40 to 250 × 10^{-6} SI units and from 1 × 10^{-3} to 10–20 × 10^{-3} A/m, respectively.

Thermal magnetic cleaning (see Fig. 3) made it possible to distinguish either two components of magnetization, low-temperature and high-temperature, or, in some cases, three components of magnetization, low-temperature, medium-temperature, and high-temperature. The first, unstable low-temperature component of the magnetization is usually destroyed at 200–250°C and corresponds to the direction of the modern magnetic field. The second, medium-temperature component is close to the paleomagnetic directions determined in the trap intrusions of the Moyero River valley [10] and, apparently, has a Permian to Triassic age. The third, high-temperature stable characteristic component of magnetization is traced up to temperatures of 660–680°C and is probably associated with hematite.

The characteristic magnetization component was successfully identified in 23 samples (Table 1). In a number of samples, it was not possible to calculate confidently the direction of the characteristic magnetization due to the overlapping of the blocking temperatures of different components. The characteristic component has two polarities, forming at least nine

Table 2. Paleomagnetic poles of the Lower Paleozoic of the Siberian platform (Angara–Anabar block). Plat is the latitude of the pole, Plong is the longitude of the pole, A95 is the radius of the confidence circle

No.	Object, area, coordinates	*N*	Plat, deg	Plong, deg	A95, deg	Source
	Middle Cambrian (513–499 Ma ago)					
1	Kulyumbe River, 68.0° N; 88.8° E		–41.9	136.2	2.4	[17]
2	Olenyok River, 71.0° N; 122.5° E		–36.4	139.6	3.8	[18]
3	Khorbusuonka River, 71.5° N; 123.8° E		–41.6	141.8	2.0	[14]
	Average	3	–40.0	139.2	5.7	Present work
	Upper Cambrian (499–488 Ma ago)					
1	Kulyumbe River, 68.0° N; 88.8° E		–36.1	130.7	2.7	[19]
2	Moyero River, 67.5° N; λ = 104.0° E		–37.4	138.4	4.6	[2]
3	Chopko River, 69.4° N; λ = 88.6° E		–39.0	138.9	4.7	[13]
	Average	3	–37.6	136.0	6.0	Present work
	Tremadocian (488–479 Ma ago)					
	Average	4	–37.9	136.4	6.9	[13]
	Arenig (~Floian and Lower Dapingian, 479–470 Ma ago)					
	Average	3	–35.2	154.9	8.5	[13]
	Llanvirn (~Upper Dapingian–Middle Darriwillian, 470–465 Ma ago)					
	Average	3	–31.6	153.7	4.7	[13]
	Llandeilo (~Upper Darriwilian, 465–461 Ma ago)					
	Average	3	–25.2	156.2	8.5	[13]
	Caradoc–Ashgill (~Sandbian and Katian, 461–446 Ma ago)					
1	Rozhkova River (middle reaches of the Angara River, Chertovskii and Baksanskii horizons), 58.5° N; 99.8° E		–29.5	140.2	6.4	[3]
2	Moyero River, 67.5° N; 104.0° E (Dolborskii Horizon)		–13.9	124.1	5.9	[2]
3	Moyero River, 67.5° N; 104.0° E (Chertovskii Horizon)		–15.7	148.6	5.7	Present work
	Average	3	–20.0	137.5	22.6	Present work

magnetopolar intervals within the horizon. The mean directions of forward and reverse polarity, however, do not pass the inversion test ($\gamma/\gamma c$ = 25.4/12.2 [11]), which is probably due to some contamination of the characteristic component with trap magnetization. Since the effect of contamination is opposite for samples with direct and reverse polarity, averaging their directions significantly reduces the effect of superimposed components up to its complete compensation. Thus, there is reason to believe that the calculated average direction is slightly offset from its true value.

The following arguments support the primacy of the characteristic magnetization identified: (1) consistency with the more ancient Ordovician directions and the proximity of the corresponding paleomagnetic pole to the pole of rocks of the same age from the outcrops of the valley of the middle reaches of the Angara River [2, 3]; (2) the presence of both direct and reverse polarity; and (3) the difference from the poles identified from younger formations of the Siberian platform and, in particular, from the pole of the Permian and Triassic trap rocks. The latter, due to their close location to the section studied, could well be a source of the rock magnetization reversal.

DISCUSSION

The paleomagnetic direction identified in the Chertovskii Horizon of the Moyero River section was used to calculate the paleomagnetic pole (see Tables 1 and 2). Together with the poles of the Dolborskii Horizon, which are close in age [2], and the Moyero River and Chertovskii–Baksanskii horizons of the Ordovician in the middle reaches of the Angara River [3], this pole can be used to determine the position of the middle paleomagnetic pole of the Upper Ordovician of the Siberian platform with a conditional age of ~450 Ma [12]. We took the middle poles for Tremadocian, Arenig, Llanvirn, and Llandaleo from [13]; to calculate the middle poles of the Middle and Upper

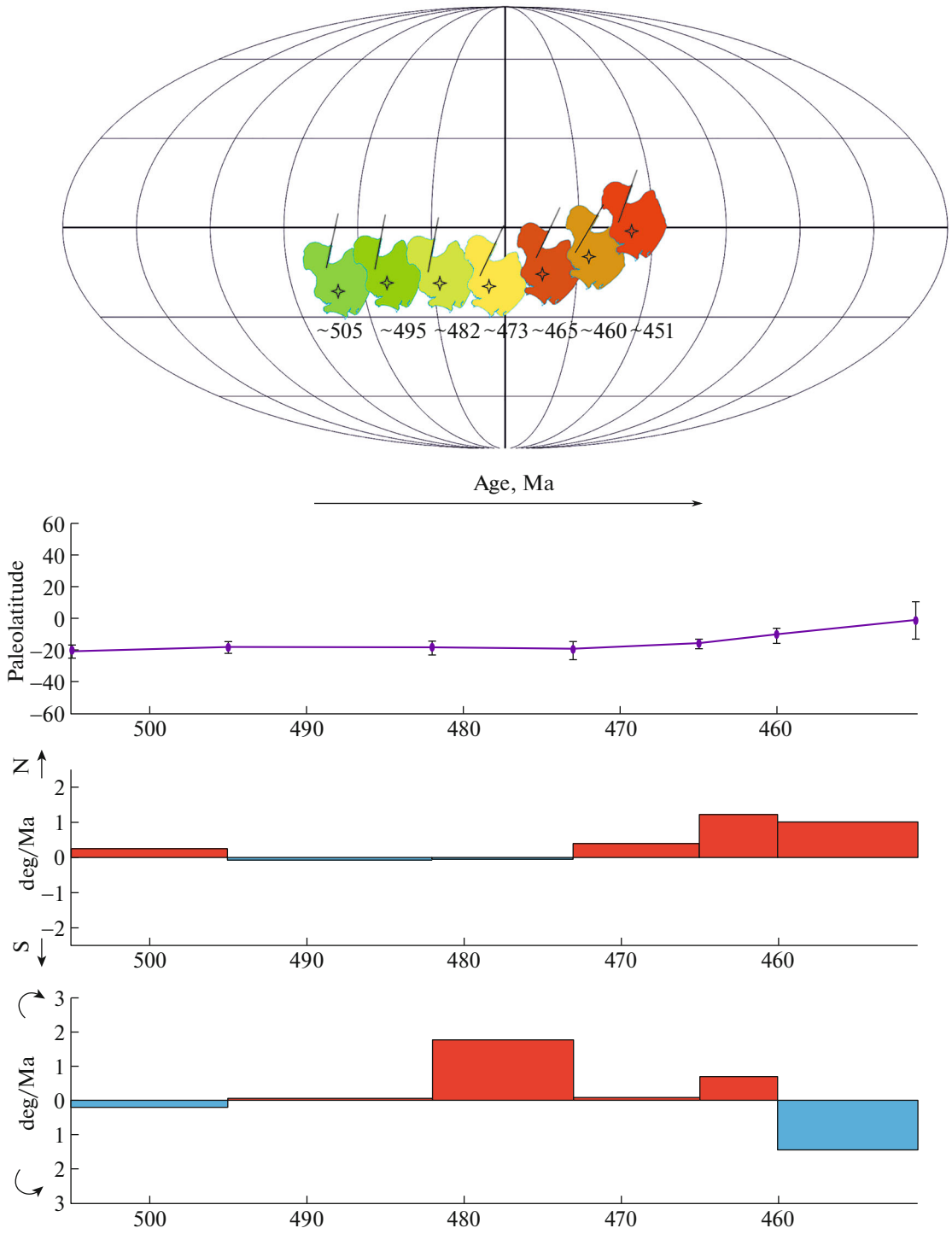


Fig. 4. Reconstruction of the paleogeographic position of the Siberian platform (top) and calculation of the parameters of its drift (bottom) in the time interval of ~505–450 Ma. In view of the paleolongitudinal uncertainty of the paleomagnetic method, the longitudinal position of the Siberian platform is arbitrary.

Cambrian, we used the recently published paleomagnetic poles [14, 15] in addition to previously used poles [13].

In total, this work presents seven high-quality poles obtained from different regions of the Siberian platform for the time interval of ~505–450 Ma. We deliberately did not use the available individual data on the Aldan Block of the Siberian platform (e.g., [16]), since in this case it would have been necessary to make an adjustment for the Late Paleozoic relative rotation of

the Angara–Anabar and Aldan blocks, which may lead to a decrease in the accuracy of the calculated poles. Note that the paleomagnetic characteristic of the Cambrian and Ordovician of the Siberian Platform is currently the best of those available for all ancient platforms and large continental blocks.

The poles obtained allow the paleogeographic position of the Siberian platform to be determined with great accuracy and its drift to be followed for over ~50 Ma from the Middle Cambrian to the middle of the Upper Ordovician.

Figure 4 shows that from the middle Cambrian (~505 Ma ago) to the end of the Early Ordovician (~473 Ma ago), the northern (in modern coordinates) part of the platform was located in the southern tropical latitudes, while the southern (in modern coordinates) part of the platform was located in the southern near-equatorial latitudes. At the same time, the modern southern part of the platform was turned towards the north. The rate of latitudinal movement of the platform at this time was low and did not exceed 0.3° per 1 Ma. By the Darrywilian time (Middle Ordovician), the latitudinal drift velocity of the Siberian Platform had increased significantly; it began to move rapidly northward at a speed of more than 1° per million years, and by the middle of the Late Ordovician, the middle part of the platform was at the equator. At that time the platform experienced only minor rotations relative to the meridian, which slightly accelerated near the Tremadocian–Floian and Darriwilian–Sandbian boundaries.

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